

If attained, resource optimisation can lead to a significant and sustainable reduction in the project duration and the number of design iterations needed to identify bugs or defects

because the engineers are able to optimise only at the sub-system or component level," explains Arun Mulpur, manager, signal processing and communications product marketing and communications, electronics, and semiconductors industry marketing, MathWorks, Inc., USA.

If the design is not validated. Most problems occur because design choices haven't been validated earlier enough. As a result, design errors or inconsistencies remain undetected until late in the design cycle, and lead to significant delays due to rework and repetition of major phases of the system's design.

Strategies, tools and best practices

Experts believe that both for engineers as well as engineering managers, design optimisation and simulation tools can be of great help in putting the safety net on to prevent resource-pilferage, and help in achieving and sustaining frugality along many dimensions. Here are a few best practices that can help.

Adopt an ideal design optimisation process. "To improve the efficiencies of the design, both design engineers and engineering managers need to make the reduction of multiple design iterations their absolute priority, and adopt model-based design for design-flows. In order to reduce the impact of incomplete or ambiguous design specifications, managers must insist on utilising executable specifications rather than paper specifications," advises Mulpur.

"Apart from this, in order to get their designs right the first time and reduce the number of iterations, managers need to utilise the executable

specifications by enabling continuous verification process," suggests Rao.

"The fundamental idea behind conducting an early validation of design is that it eliminates the risks associated with faulty designs. It also makes it possible to integrate the different system elements and test the system as part of the development process, instead of a big bang integration and testing at the end of the development cycle when resources, time, budget and funding have already been consumed," cautions Michel Genard, vice president, product strategy and marketing—Simics, Wind River.

Know the architecture, inside out. "With advances in technology nodes, FPGA architectures are changed and optimised so as to boost the performance, decrease power consumption and reduce resource wastage. Before designing for a device, the engineers should be well aware of the architecture nuances so that they could make the best use of their target device," opines Jain. FPGA companies publish recommendations on design practices. Following these recommendations also help in getting an optimised design.

Choose the correct design tools. In addition to formulating the correct design evaluation and review strategy, optimising resources through specific design optimisation tools also proves useful. Doing this enables design engineers to validate and verify the architecture, make sound implementation choices, and debug hardware or software flaws.

"In general terms, to be useful, design tools should both provide flexibility and interoperability," Jain adds. While flexibility enables the engineers to either use command lines (or script) or user interfaces, interoperability provides the connection points with other tools or design flow to make global collaboration possible. These attributes should define the selection and usability of design tools.

Genard, however, makes an interesting point with respect to design tools' efficacy. He says, "Design tools

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